

Digital Banking BI & Data Warehouse

Personal Finance and Spending Analytics

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An end-to-end business intelligence project that converts synthetic digital-banking transaction data from a normalized OLTP model into an analytics-ready snowflake data warehouse using MySQL, KNIME ETL workflows, SQL aggregation views, and BI reporting.

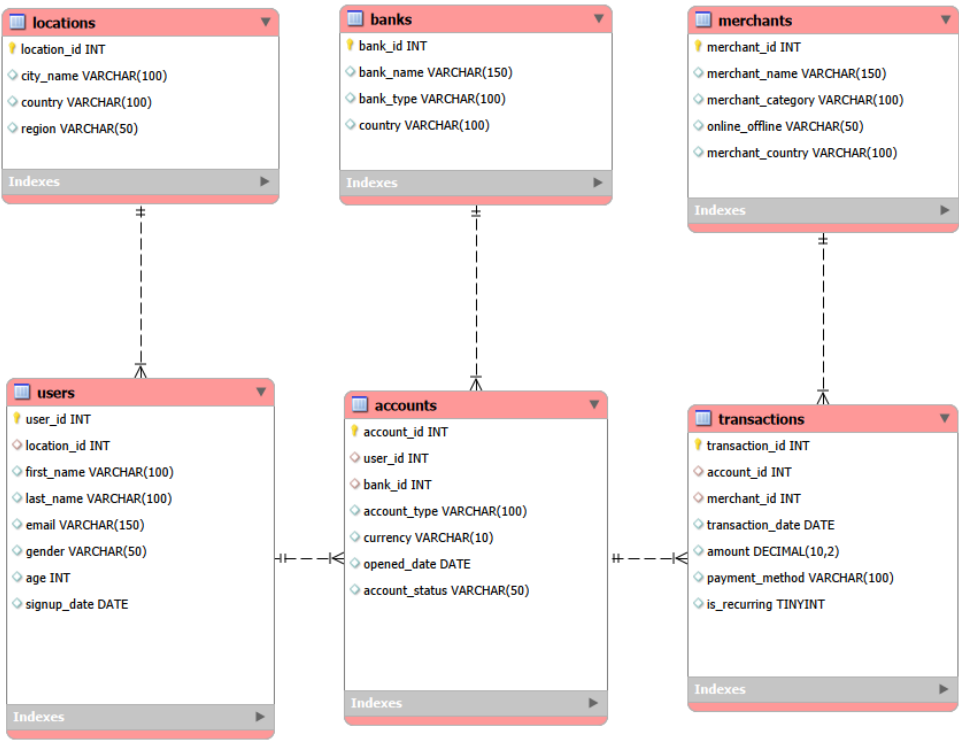


Figure 1. Normalized OLTP source model for users, accounts, banks, merchants, locations, and transactions.

1. Data architecture

The operational source captures one transaction at a time and uses foreign-key relationships across users, accounts, banks, merchants, and locations. For analytical workloads, the project separates operational storage from reporting by introducing a dedicated data warehouse.

Snowflake warehouse

The warehouse is centered on **fact_transaction** with a grain of one row per transaction. Core dimensions include date, user, account, merchant, and payment method. User, account, and merchant dimensions are further normalized through location, bank, merchant-category, and sector tables.

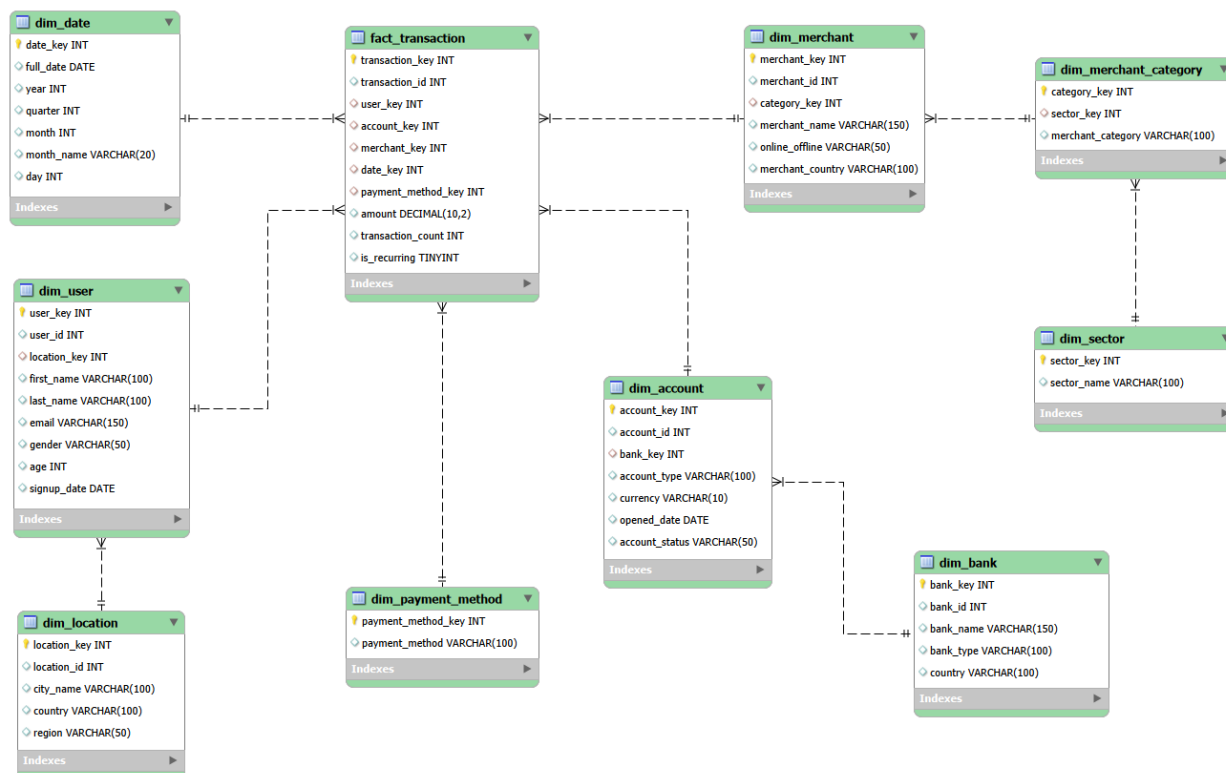


Figure 2. Snowflake data warehouse design.

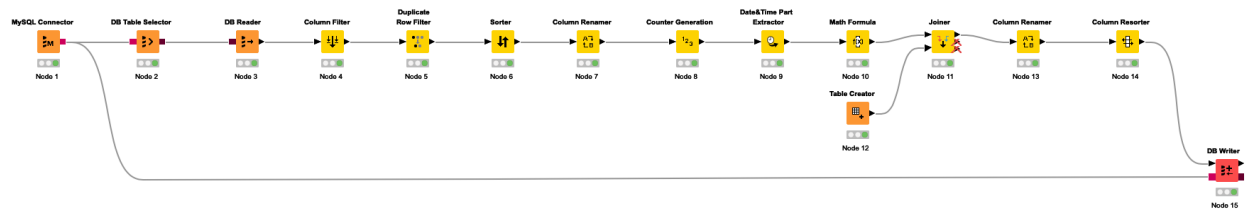
Warehouse element	Role
fact_transaction	Transaction-level fact table
amount	Primary spending measure
transaction_count	Supports frequency analysis
is_recurring	Distinguishes recurring and one-time spend

2. ETL workflow

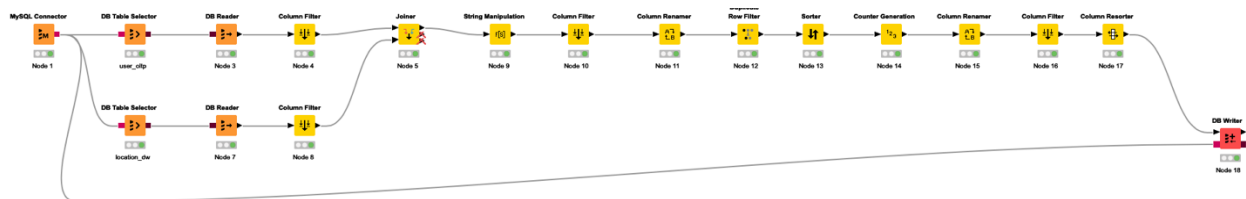
KNIME Analytics Platform is used to move data from the OLTP schema into the warehouse. The workflows perform extraction, column filtering and cleaning, lookup joins, surrogate-key generation, deduplication, sorting, transformation, and final database writes.

Representative transformations

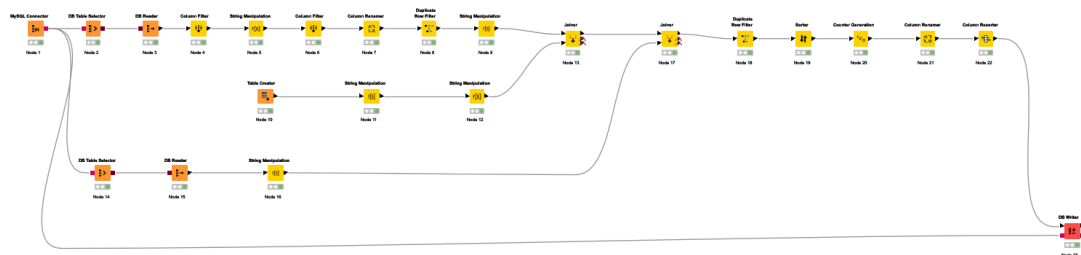
Date dimension - Creates date_key values and derives calendar attributes from transaction dates.



User dimension - Maps OLTP users to warehouse location keys, cleans fields, and generates user_key values.



Merchant-category dimension - Cleans merchant categories, maps them to broader business sectors, and retrieves sector surrogate keys.



3. SQL analytics and BI reporting

A MySQL view summarizes monthly spending by merchant sector for 2024. It joins the transaction fact table to date, merchant, merchant-category, and sector dimensions, then reports transaction count, total spending, and average transaction amount.

The reporting workflow reads the view into KNIME, aggregates yearly sector totals, calculates average transaction amount, sorts sectors by total spend, and produces a bar chart.

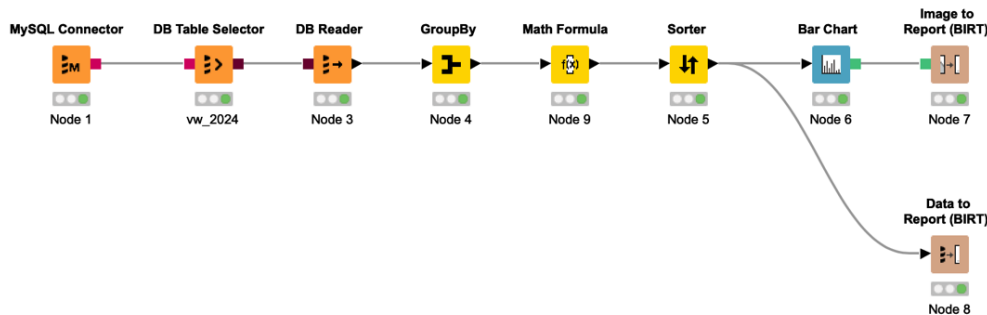


Figure 3. KNIME reporting workflow.

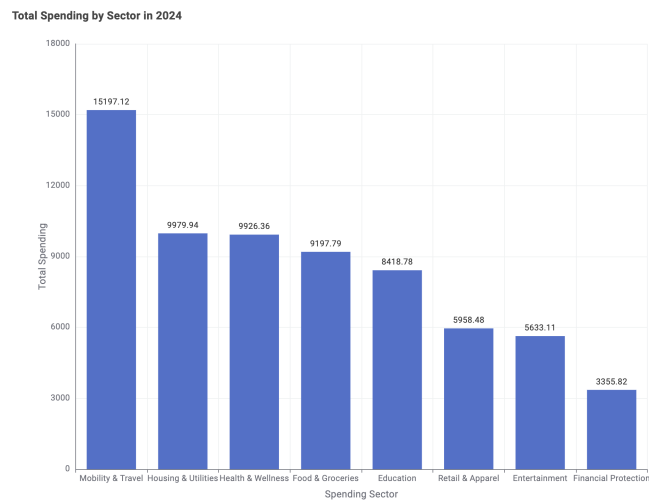


Figure 4. Total spending by sector in 2024.

Key result: Mobility & Travel shows the highest total spending in 2024, followed by Housing & Utilities, Health & Wellness, and Food & Groceries. Financial Protection has the lowest total spending among the displayed sectors.

4. Skills demonstrated

Area	Demonstrated capability
Data modelling	Normalized OLTP schema; fact/dimension and snowflake warehouse design
ETL	KNIME extraction, cleaning, joins, surrogate keys, transformations, DB writes
SQL	Multi-table aggregation view for monthly sector spending
BI reporting	Sector-level aggregation and bar-chart reporting
Data governance	Synthetic source data; no real customer information in the project

Provenance note: The underlying technical artefacts and results originated in a collaborative academic project. This report is a cleaned portfolio presentation and does not claim sole authorship of the original group work.